



IES E-CAMPUS: MASTER PRODUCT REQUIREMENT DOCUMENT (PRD)

Unified System Architecture, Functional Specifications & Cross-Role Operational Workflows

Document Status: Final Approved Baseline (Enterprise Academic Edition)
Target Institution: IES College of Technology / IES University, Bhopal (RGPV & AICTE Approved)
Product Name: IES E-Campus ERP Ecosystem
Technology Foundation: Flutter Multi-Platform (Android / iOS / Web) + Cloud Firestore (Zero Cloud Storage Cost Architecture)
Version: 2.4.0 (Production Release)



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1. Executive Summary & Product Vision

1.1 The Core Problem

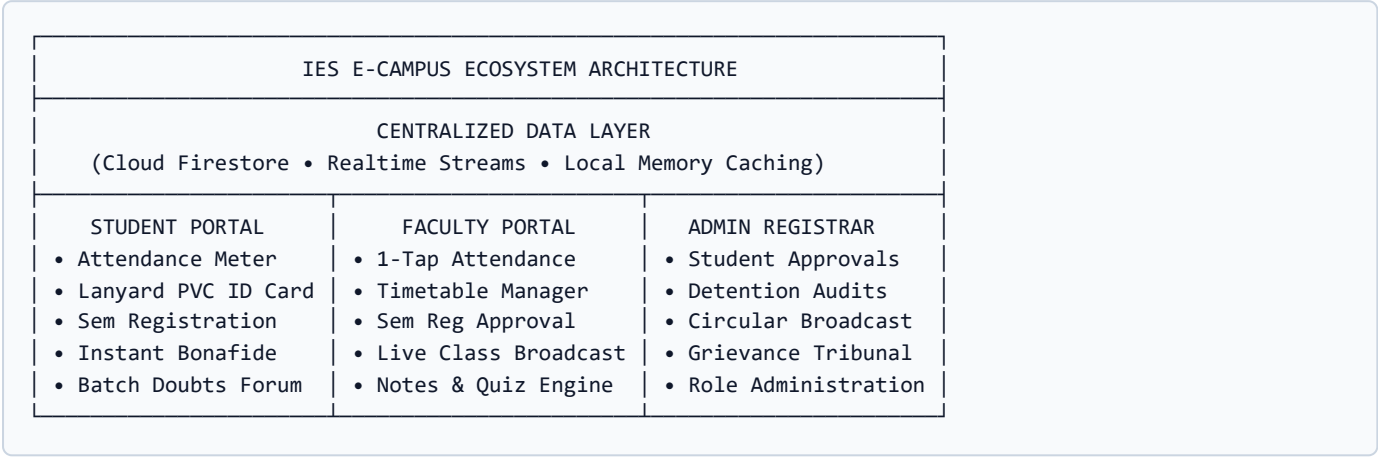
Most technical universities and engineering institutions across India still rely on fragmented, paper-based administrative workflows:

- **Attendance:** Physical paper register books waste 15 minutes of every lecture; students shout proxies; manual calculation of monthly percentages leads to errors.
- **Certificates:** Bonafide certificates and semester registration forms take 3 to 7 days of routing across clerk, HOD, and principal offices.
- **Identity Cards:** Simple horizontal paper/plastic cards without security stamps or lanyards fail basic campus gate

- verification.
- **Class Schedules:** Stagnant notice board paper printouts fail to inform students when a professor is on leave or an extra lab is scheduled.
 - **Academics:** Study notes are shared haphazardly on WhatsApp groups where messages get deleted, personal phone numbers are exposed, and files are lost.

1.2 The IES E-Campus Solution

IES E-Campus is a unified, paperless Enterprise Campus Operating System designed ground-up for Indian engineering institutions (specifically aligned with **IES College of Technology, Bhopal**, affiliated with **RGPV** and governed by **AICTE** norms).



2. Stakeholder Personas & Cross-Role Interaction Matrix

2.1 Persona 1: The Student (Candidate Rahul Sharma)

- **Role:** 3rd Year (5th Semester) B.Tech in Computer Science & Engineering, Section A.
- **Goals:** Stay above the 75% attendance threshold, download study materials without cost, obtain instant bonafide certificates for state scholarships (MPTAAS), and stay updated on daily timetable changes.
- **Pain Points:** Long administrative queues, unexpected detention lists, losing track of assignment deadlines.

2.2 Persona 2: The Faculty Member (Prof. Mohit Donawat)

- **Role:** Head of Department (H.O.D.) / Associate Professor, Computer Science.
- **Goals:** Conduct lectures without wasting time on paper roll calls, schedule substitute teachers when on leave, review semester registration applications in 1 tap, and moderate class doubts safely.
- **Pain Points:** Paper registers, manual attendance percentage calculations, students submitting incomplete registration forms.

2.3 Persona 3: The Administrator / Registrar (Dr. R.K. Saxena)

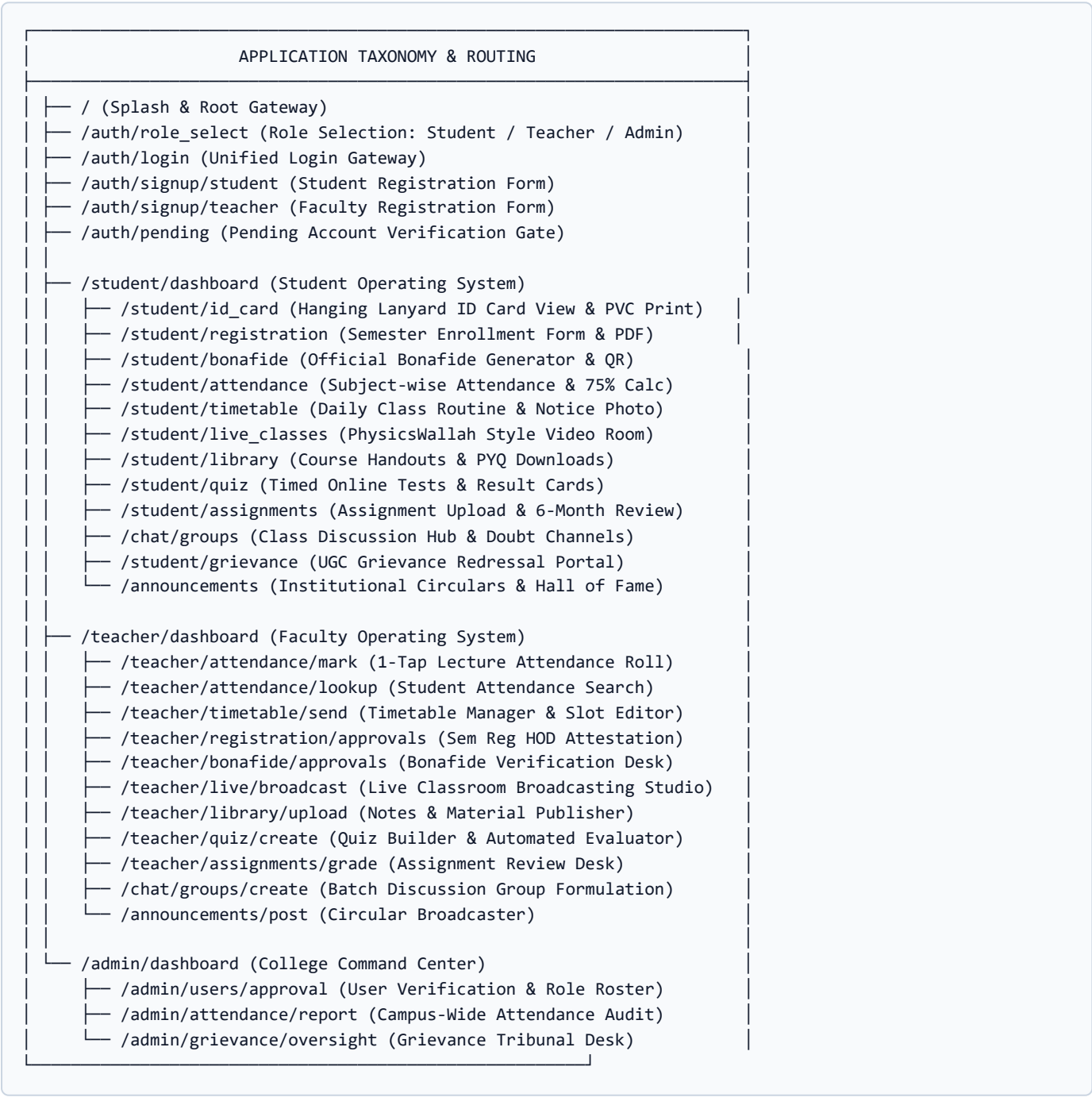
- **Role:** College Registrar & Chief Examination Officer.
- **Goals:** Ensure institutional compliance with RGPV/AICTE guidelines, audit detention lists before issuing university exam admit cards, monitor campus grievances, and prevent identity fraud.
- **Pain Points:** Forged physical bonafides, lack of real-time attendance transparency, chaotic grievance records during AICTE inspections.

2.4 Cross-Role Cascading Interaction Matrix

Initiating Action	Triggered By	Immediate Impact on Student	Immediate Impact on Faculty	Immediate Impact on Admin
Mark Class Attendance	Faculty Member	Student's attendance % updates in <1 sec; status badge changes to Green/Amber/Red.	Lecture record stored in faculty history; statistics update.	College-wide daily attendance KPI updates on Master Dashboard.
Update Daily	Faculty	Student daily routine highlights	Schedule conflicts	Real-time classroom

Initiating Action	Triggered By	Immediate Impact on Student	Immediate Impact on Faculty	Immediate Impact on Admin
Timetable	Member	substitute period with teacher name and room.	prevented; substitutes clearly noted.	utilization visible to management.
Submit Sem Registration	Student	Receives auto-populated PDF with unique ref number; waits for HOD sign-off.	HOD receives applicant card in review queue; inspects SGPA/backlogs.	Registration counter increments towards 100% target; fee status tracked.
Attest Bonafide Certificate	Faculty / HOD	Student mobile app receives instant download link for sealed PDF with QR code.	Application marked resolved; audit trail recorded.	Validated bonafide logged with unique reference number for scholarship verification.
Broadcast Urgent Notice	Administrator	High-priority banner and push alert display on student devices instantly.	Faculty notified of institutional decisions (e.g. holiday, exams).	Complete audit trail of message dispatch timestamp.

3. Unified Information Architecture & Screen Navigation Hierarchy



4. Deep Functional Specifications (12 Subsystems)

4.1. Authentication & Two-Tier Security Gate

- **Input Data:** Full Name, University Roll Number, RGPV Enrollment Number, Department, Year, Semester, Section, Email, Password, Profile Photo.
- **Processing:**
- Writes new document to `users` collection with `isApproved = false`.
- Student app locks on `PendingApprovalScreen`.
- Dispatches alert notification to faculty dashboard.
- Faculty inspects roll number and taps **Approve**.
- Database updates `isApproved = true`, unlocking student app automatically in real time.
- **Output:** Authenticated session with role-based routing.

4.2. Hanging Lanyard PVC Identity Card Engine

- **Visual Design:** Vertical ID card with a top metallic lanyard punch hole band, institution crest, golden session ribbon, passport photo with holographic security border, credentials table, crimson circular seal, registrar signature, and barcode.
- **Back Side:** Emergency contacts, blood group, guardian details, return campus address, and 5-point institutional discipline rules.
- **Export Formats:**
- **Gallery PNG:** Flutter `RepaintBoundary` rasterizes pixels at **300+ DPI** directly to device gallery.
- **Print-Ready A4 PVC PDF:** Renders Front and Back side-by-side with alignment crop markers for PVC printing machines.

4.3. Official Semester Registration & Course Enrollment System

- **Real-College Mandate:** Matches standard Indian university semester promotion forms (Part A to Part E).
- **Required Inputs:**
- Auto-populated: Student Name, Enrollment, Roll No, Branch, Year, Father's Name, Phone, Email.
- Student Enters: Target Semester, Academic Session, Previous Semester SGPA, Cumulative CGPA, Backlog/ATKT flag (with subject codes), Co-curricular Achievements, Elective Course Choices, Fee Receipt Ref Number.
- Legal Undertaking: Mandatory agreement to maintain 75% attendance and adhere to anti-ragging policies.
- **Approval Workflow:**
- Form appears in HOD's review desk.
- HOD inspects credentials and taps **"Approve & Sign"**.
- Digital HOD signature and crimson red college seal are applied.
- **Output:** Official print-ready A4 PDF document (`IES/REG/[YEAR]/[ENROLL]`).

4.4. Digital Bonafide Certificate Generator & Attestation Desk

- **Purpose Selection:** Templates for NSP/MPTAAS Scholarships, Bank Education Loans, Passport/Visa, Bus/Train Pass, and Custom Purposes.
- **Attestation Elements:**
- Phrasing: *"This is to certify that [Name], Son of [Father's Name]..."*
- Attested student passport photograph.
- Crimson red double-ring college seal stamp with 12 stars.
- Vector Registrar / Principal signature block.
- Dynamic verification QR code encoding candidate UID, enrollment number, and timestamp.
- Authoritative guarantee banner confirming validity for all government and banking submissions.
- **Output:** Instantly downloadable and shareable A4 PDF.

4.5. Classroom Attendance Engine & 75% RGPV Criteria Radar

- **Teacher Workflow:**
 - Select Branch, Semester, Section, Subject, and Period.
 - Tap "**Mark All Present**" (turns 60 students green in 1 second).
 - Tap absent students to turn them red.
 - Tap "**Submit Attendance**" (takes less than 20 seconds total).
- **Mathematical Formulas:**
- **Live Attendance Percentage:**
$$\text{Percentage} = \left(\frac{\text{Total Attended}}{\text{Total Held}} \right) \times 100$$
- **Smart Prediction Formula (When Below 75%):**
$$\text{Classes Needed} = \left\lceil \frac{0.75 \times \text{Total Held} - \text{Total Attended}}{0.25} \right\rceil$$
- **Admin Audit:**
 - Cross-department detention radar.
 - 1-click export of defaulter lists for RGPV examination admit card withholding.

4.6. Master Timetable Architecture (Weekly Default vs Today's Substitute)

- **Class Filtering:** Specific timetable for every Branch, Year, Semester, and Section.
- **Dual Mode:**
- **Weekly Master Schedule (Daily Default):** Permanent Monday-to-Saturday routine.
- **Today's Reschedule / Substitute:** Day-specific modifications (e.g. substitute professor, rescheduled timing, or extra lab).
- **Auto-Fill AICTE/RGPV Curriculum:** 1-click button populates core semester subjects automatically (OS, DBMS, CN, SE, Labs), saving hours of manual data entry.
- **Period Cards:** Displays Period Number (1 to 8), Time range, Subject name, Room/Lab, and Faculty in-charge with Edit/Delete buttons.
- **Notice Board Photo:** Optional camera upload of physical department notice board.

4.7. Live Interactive Classroom Studio (PhysicsWallah / Unacademy Style)

- **Broadcasting Engine:** Embedded HD video streaming player supporting live lectures and recorded archives.
- **Live Doubts Chat:** Real-time Q&A feed between students and faculty during the lecture.
- **Role Badges:** Faculty responses stand out in golden cards with verified professor tags.
- **Lecture Handouts:** Attached slide presentations (PPT/PDF) available for download below the video player.

4.8. Digital Library & Academic Handouts Repository

- **Categorization:** Filtered by Department, Year, and Subject.
- **Document Types:** Lecture Notes, Syllabus Copy, Lab Manuals, Previous Year Question Papers (PYQs with solutions).
- **In-App Viewer & Offline Download:** Students can read documents inside the app or download them for offline study.

4.9. Timed Online Quiz & Automated Examination Engine

- **Faculty Builder:** Create multiple-choice questions with 4 options, correct answer key, and countdown timer.
- **Student Testing:** Clean test-taking interface with timer, answer tracking, and review status.
- **Instant Auto-Grading:** Grades are calculated immediately upon submission, with scorecards and batch leaderboard rankings.

4.10. Assignment Publishing & 6-Month Review Archive Desk

- **Faculty Desk:** Post assignment problems with deadlines and reference sheets; monitor real-time

submission counter (X / Y Students Submitted).

- **Student Portal:** Upload handwritten scans or typed PDF reports with automatic timestamp verification.
- **6-Month Archive:** Retains graded submissions, scores, and faculty remarks for half a year for academic audit compliance.

4.11. Anti-Tamper Class Discussion Groups & Batch Mentorship

- **Targeted Batch Groups:** Automatically channels students into groups matching their Branch, Year, Semester, and Section.
- **Canonical Matching:** Intelligently matches abbreviations (CSE $\rightarrow$ Computer Science & Engineering , 3rd Year $\rightarrow$ 3 , 5th Sem $\rightarrow$ 5).
- **Anti-Tamper Protocol:**
- **No Deletion:** Messages cannot be deleted once posted, maintaining academic decorum.
- **60-Second Edit Window:** Typos can only be corrected within 1 minute, displaying an [edited] tag.
- **Instant Messaging:** Optimistic local rendering delivers messages in <1 ms.

4.12. Statutory Grievance Redressal & Anti-Ragging Cell

- **Compliance:** Adheres to mandatory UGC (Grievance Redressal) Regulations and AICTE Anti-Ragging Directives.
- **Confidential Submission:** Option to file anonymously or with verified identity.
- **Lifecycle Tracking:** SUBMITTED $\rightarrow$ ASSIGNED $\rightarrow$ INVESTIGATION $\rightarrow$ ACTION TAKEN REPORT (ATR) $\rightarrow$ RESOLVED .

5. UI/UX Design System, Color Tokens & Typography

5.1 Color Tokens

- **Background:** #0F0C20 (Deep Space Midnight)
- **Surface:** #1A1635 (Sleek Dark Glassmorphism)
- **Surface Variant:** #252048 (Elevated Container)
- **Primary Brand:** #1E1B4B (Royal Academic Navy)
- **Secondary Accent:** #F59E0B (Academic Gold / Amber)
- **Success / Safe Zone:** #10B981 (Emerald Green)
- **Warning / Condonation Zone:** #F59E0B (Amber Orange)
- **Error / Detention Zone:** #EF4444 (Crimson Red)
- **Info / Engineering Blue:** #38BDF8 (Sky Cyan)
- **Card Gradients:** Linear gradient from #1E1B4B to #2E1065 .

5.2 Typography Standards

- **Display Headings:** Bold sans-serif with slight letter spacing (letterSpacing: 0.8).
- **Body Text:** High-legibility sans-serif with calibrated line height (height: 1.4).
- **Status Chips:** 9px to 11px uppercase bold badges (fontWeight: FontWeight.w800).

6. Institutional Document Verification & Security Standards

6.1 Authentic Crimson Red College Seal Stamp

- Double-ring concentric circle design in deep maroon ink (#8B0000 / #991B1B).
- Outer perimeter: 12 decorative stars.
- Arching text: "★ IES UNIVERSITY ★" (top) and "BHOPAL (M.P.)" (bottom).
- Inner ribbon: "OFFICIAL SEAL • GOVT APPROVED" .

6.2 Registrar Signature Block

- Stylized vector cursive signature of the authorized institutional signatory.
- Designation pill: "REGISTRAR / PRINCIPAL AUTHORIZED SIGNATORY".

6.3 Security Verification QR Code Schema

Every generated certificate carries a structured JSON payload:

```
{
  "ref": "IES/BONA/2024/7819",
  "enroll": "0103CS221045",
  "name": "Rahul Sharma",
  "branch": "CSE",
  "sem": "5th Sem",
  "inst": "IES College of Technology, Bhopal",
  "issued": "2024-09-08T10:30:00Z"
}
```

7. Non-Functional Requirements, Cloud Economics & Security

7.1 Free-Tier Architecture (Zero Cloud Storage Bucket Costs)

- Large external cloud storage buckets often incur monthly bandwidth fees and CORS configuration issues.
- **IES E-Campus Innovation:** Document scans and photo attachments are compressed and encoded as **Base64 Data URIs** stored directly in Cloud Firestore.
- **Zero-Index Query Model:** Data queries use in-memory stream filtering, eliminating the need for composite indexes that can fail in unindexed Firestore databases.

7.2 Anti-Tamper Protocol

- Discussion group messages cannot be deleted.
- Edits are limited to 60 seconds with an audit tag.
- Attendance records can only be submitted by verified faculty accounts.

8. Master End-to-End User Scenarios & Cross-Role Flows

Scenario: Monday Morning Lecture & Timetable Reschedule

1. **08:15 AM (Faculty Reschedule):** Prof. Mohit Donawat has a department meeting at 10:00 AM. He opens `SendTimetableScreen`, selects *Today's Substitute*, and assigns Prof. Sharma to cover Period 2 in Computer Lab 2.
2. **08:16 AM (Student Notification):** Rahul Sharma opens his student dashboard. Period 2 displays a yellow `[ ⚡ SUBSTITUTE ]` badge indicating Prof. Sharma in Lab 2.
3. **10:05 AM (Lecture Attendance):** Prof. Sharma takes the class, opens `MarkAttendanceScreen`, taps `Mark All Present`, unchecks 2 absentees, and submits in 14 seconds.
4. **10:06 AM (Real-Time Recalculation):** Rahul's attendance meter updates to 82.4% (Green zone). The 2 absentees receive attendance warning notifications.
5. **10:45 AM (Doubt Clearing):** In `GroupChatRoomScreen`, student Priya asks a question using the quick chip " `? Ask Question`". Prof. Sharma replies with an authoritative golden `👤 FACULTY` badge.
6. **11:00 AM (Admin Oversight):** The Registrar opens the admin dashboard and sees today's CSE 3rd Year attendance logged at 96.7%, with zero paper registers required.

9. Executive Presentation Guide for Evaluators & Management

When presenting this system to college authorities, evaluators, or investors, use this structured narrative:

1. **The Hook:** "Every semester, over 2,000 students wait in queues for Bonafide certificates, semester registration,

and attendance verification. Professors waste 15 minutes of every lecture taking roll calls on paper."

2. The Demonstration:

3. **Show Student Portal:** The live circular attendance meter, instant Bonafide generator with QR code, and vertical hanging lanyard ID card.

4. **Show Faculty Portal:** The 20-second "Mark All Present" attendance roll call and the master timetable substitute manager.

5. **Show Admin Portal:** Campus-wide 75% detention radar and 1-tap parent calling/emailing.

6. **The Technical Innovation:** *"We built this entire system with zero cloud storage costs by leveraging direct Base64 Firestore persistence, eliminating recurring cloud hosting bills for the institution."*

7. **The Regulatory Compliance:** *"Every document complies with RGPV Ordinance 12, AICTE norms, and UGC anti-ragging mandates, complete with authenticated crimson seals and QR verification."*